MTE 203 – Advanced Calculus
Week 1 Tutorial

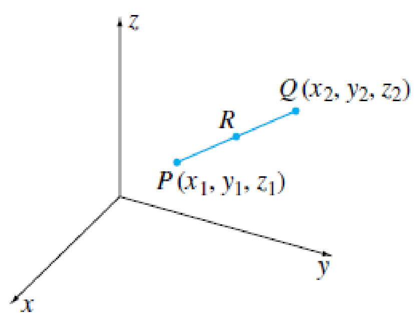
Problem 1: (Ch11.1, Prob 19)

If P and Q in the figure below have coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2), prove that if a point R divides the length of the segment PQ so that

$$\frac{|\overrightarrow{PR}|}{|\overrightarrow{RQ}|} = \frac{r_1}{r_2},$$

where r_1 and r_2 are positive integers, then the coordinates of R are,

$$x = \frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}, \quad y = \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2}, \quad z = \frac{r_1 z_2 + r_2 z_1}{r_1 + r_2}$$



Problem 2:

Find the vector form of the equation for the line in the plane $z = 3$ that makes an angle of $\frac{\pi}{6}$ rad with $\hat{\imath}$ and of $\frac{\pi}{3}$ rad with $\hat{\jmath}$. Describe the reasoning behind your answer.

Problem 3

The lines L_1 and L_2 are represented by the following parametric equations,

$$L_1: \quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t$$

$$L_2: \quad x = 2s, \quad y = 3 + s, \quad z = -3 + 4s$$

a) Write their vector form
b) Find their symmetric equations
c) Show that L_1 and L_2 do not intersect